INGB 417 EXAM GA PROJECT

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1.Executive Summary

To design a effective facility layout for WB Equipment a specific system engineering designing process was followed. Firstly the objectives and requirements were setup by matching it with WB Equipment's goal for this new facility. It is a very important aspect of the project, as it will be used at the end to evaluate the success and effectiveness of the design. The next important step is a needs analysis, and will include the analysing of production capacity and need for improvements, the equipment required according to cycle times of the product, minimising distance travelled between departments and to group important departments together, optimising space utilisation. Then the alternative designs can be made using the systematic layout planning method considering different type of layout shapes and grouping of the operational processes including machines in different ways. In this project the gap analysis was done to evaluate which design fits the requirements set in the beginning. Our final design promotes flow, includes an effective operational process by grouping the departments

in the best logical manner, ensuring the best quality control which gives WB equipment the ability to only grow from now on in their supply to the expanded market.

2.Background

Wb equipment is a wheelbarrow manufacturing company who manufactures the different parts and assembles it as a final product. They have supplied the South African market for the past 35 years. Until recently they were supplying only to the local market, until they acquired a competitor who has run into financial difficulties, who also has a network in the export market which means that they must expand their current facility. They have bought a new facility in another province, strategically placed to supply the export market. The goal of the project is to develop a new design, within the requirements given, by using systematic engineering design skills.

3.Design problem + Objectives

3.1 Problem areas to address

WB Equipment have always supplied for the local market, but now they experience an increase in their demand with almost 7 times more, as the competitor who they acquired has a network in the export market. The company's current facility is therefore not able to produce enough products to satisfy this new demand which is a big adjustment. Taking the export market's location into consideration, the current facility's location would make supplying difficult and therefore they will build a new facility that is in a strategically improved location. They now face the challenge of designing a new facility for the proposed site that will meet customer demand as they will now have much bigger orders. The acquired competitor suffered from financial difficulties and therefore it creates pressure for WB Equipment to not make the same mistakes, thus it is suggested to be secured in ensuring this newly designed facility is an improvement on the previous facility design.

3.2 Objectives

The objective of the project is to design a facility that will improve operational efficiency to reduce costs and increase productivity, but also mainly focuses on expanding production capabilities so that WB Equipment can continuously improve their supply of products and meet the new demand of the local and now export market.

The objectives of the facility design will reflect on WB Equipment's vision and mission to efficient operational facility to meet their demand in the following ways:

- Maximise profitability

The facility design will maximise profitability by reducing costs, improving operational efficiency, optimising material and operator flow and supply chain, ensuring quality control and elimination of waste disposal.

- Effective use of space
  Designing a facility that will make efficient use of space requirements in this 4HA providing for the manufacturing area, inventory space for raw materials, finished goods as well as Work-In-Progress storage and the supporting facilities.

- Facilitate design for future growth
  Design the facility while accommodating for potential future expansion leaving 30% of the land available for growth without disrupting current operations.

- Optimise supply chain
  Improve the efficiency of the supply chain including the best layouts of designated receiving and dispatch areas that will minimise material handling time and costs.

- Customer satisfaction
  We will continuously improve by having quality inspections of the raw materials, during the manufacturing as well as the final product to minimise and correct defects. Employees will have ongoing training programs to make sure their skills are up to standard, and equipment and tools will receive regular maintenance.

- Employee satisfaction
  Implement a safe working environment for our employees adhering to all regulations such as emergency and health protocols.

- Waste disposal
  Design a recycling system to minimise our environmental footprint by making use of our scrap for a second time.

4.Requirements

The following requirements for the facility design have been established in accordance with the objectives of the organisation. The requirements will be used as criteria to evaluate the alternative designs, ensuring to decide on the best design.

1. Production Capacity

A production capacity analysis will be done as well as evaluating if the layout promotes effective space for the required machines and personnel to determine if the facility will be able to meet the future product demand

2. Equipment and machinery
   We should have enough effective working machinery and equipment that will produce our required future demand of products and therefore it will be inspected on a regular basis to make sure it is working on a quality standard and if necessary, repairs will be done as soon as possible to prevent downtime in our production.

3. Space utilisation
   The size of the manufacturing plant must be 4 HA. The design should promote the utilisation of space in the facility by ensuring logical flow between departments and processes and prioritising space allocation of activities based on frequency and importance. The design should distribute space allocations for the storage areas, receiving and dispatch areas, manufacturing processes including all the machines required, other supporting departments for staff while providing 30 % space for future expansion within the 4 HA provided.

4. Promote efficient flow
   The facility layout must be designed in such a way that the different workstations of the manufacturing, storage areas and other supporting departments are placed in the best combination that will optimise flow of materials and workers to optimise bottlenecks and ineffective use of space. An activity relationship chart will be used to ensure this efficient flow.

5. Quality Control
   There will be quality inspection points of the raw material, throughout the manufacturing as well as the final products to ensure the product standards are continuously up to standard and to minimise defects by designing a layout that provides operational space for these quality assurance systems and that will be able to fix these possible defects as soon as possible.

6. Future expansion
   The design will ensure that there is enough space for a potential 30% expansion by including extra 30% space in all the departments inside the

building and making use of an open plan design to promote flexibility for changes to be made at any time.

7. Safety regulations
   The design will ensure an implementation of a fire safety system, enough air ventilation, as well as necessary utilities such as water, electricity and plumbing will be certified and receiving regular maintenance. All equipment and machinery will be inspected regularly, and repairs will be done in accordance, to ensure a safe working environment for all the workers. A security system will be implemented to ensure safety of all employees and control over the whole plant.

8. Promotes supply chain
   Improve the efficiency of the supply chain by designing operational areas for receiving and dispatch areas that optimises a logical flow of the materials, products and the trucks and providing necessary facilities for truck drivers to prevent unnecessary delays in this part of the operation. An implementation plan will also be included for WB Equipment's own fleet taking large orders to distribution areas.

5.Assumptions and Constraints

To do a complete needs analysis of the project we will make the following assumptions and create the following constraints.

5.1 Assumptions

5.1.1 Labour assumptions

- Assume 240 working days per annum/year.
- Assume 5 working days in a week.
- Assume 4 weeks in a month.
- Assume 9 working hours per day with an hour downtime (lunch and tea breaks).

5.1.2 Scrap assumptions

- The off cuts of the steel sheets will be used to shape the brackets by the milling machine.
- Assume the rest of the plant as a 0% scrap rate.

5.1.3 Operational assumptions

- Assume 2 weeks inventory space for raw materials.

- Assume 1 week inventory space for finished goods.
- Assume a large order includes more than 100 wheelbarrows and a small order less than 100.
- Assume WB Equipment use their own fleet for large orders and supplier vehicles for small orders.
- Assume raw materials for plastic handle caps as plastic pallets.
- Assume raw material for brackets as the off cuts of the steel sheets when shaping the tub.
- Assume the steel tube's size as 3.6 m x 30 mm thick.
- Assume the wheels are purchased from a supplier.
- Assume the bolt size as 50mm (L) x 10mm (diameter), the washers as 20mm diameter and the nuts as 15mm (W) x 10mm (H)
- Assume the bolt size as a cylinder, the washers as a circle and the nuts as a rectangular prism to simplify the volume calculations of each.
- Assume a standard of 80 % utilisation of all the machines to account for possible downtime so that it will not affect the process with possible bottlenecks.
- We will have 5 of our own trucks for large orders.
- We will set up solar panels in the parking area to help with electricity suppliance and our water supply systems will be maintained regularly by a plumbing company we have a contract with. They will also have to approve the water system in implementing the design.
- The milling machine used to produce the brackets is a multifunction machine that can shape all the types of brackets.
- Assume the density of steel as 7850 kg/m^3
- Assume the dimensions of the steel sheets as 2.5m x 1.2 m.
- Assume a timber pallet will be used for steel sheets with dimensions 3m x 1.2m that weighs 47 kg and can handle heavy duty loads, of max 1500 kg.
- Assume we will paint everything with a single coat of paint.
- Assume 1 Liter of paint covers approximately 10 m^2.
- Assume the plastic handlebar size as 30 mm diameter, 125 mm long with a weight of 0.15kg per piece.
- Assume the wheel rubber bracket comes with each wheel that is bought.
- Assume the daily demand of plastic pellets will be put into the moulding machine's hopper at once in the beginning of the day so that it will not take up extra space.
- Assume operators will be packing and sealing the finished wheelbarrow into the boxes.
- Assume the wrapping for the final product is Aerothene Foam wrap. It is 1,5m x 200m of wrapping on one roll. The dimensions of one roll are therefore 1,5 x 1 m.

- Assume the forklifts for the raw material storage and finished goods are parked separately in these storage areas.
- Assume 4m x 4m floor space is given for a robot arm.
- Assume a conveyer belt of will be used to move the off cuts from the trimming machine to the milling machine which serves as WIP.
- Assume the milling machine will automatically be fed with the raw materials (off cuts).
- Assume our own fleet of trucks sizes as 13.5 x 2.45 x 3 m with a trailer of 8m x 2.45 x 3m to make sure it can hold orders of 100 and more orders.
- Assume we will have 5 of our own trucks for large orders.
- Assume a 20% aisle space and extra 20% forklift space for the whole plant.
- Assume the quality inspection points will be done by the supervisors appointed and it will be done while the process is done so there are no bottlenecks in these parts.

5.2 Constraints

5.2.1 Stock constraints

- Inventory space for raw materials must provide for 2 week's stock.
- Inventory space for finished goods must provide for 1 week's stock.

5.2.2 Size constraints

- The size of the manufacturing plant must be within 4 HA.
- The facility must leave 30 % space available for possible future expansion needs.
- The weight of the steel sheets is calculated as 113.04 kg.
- Our own fleet of trucks sizes are 13.5 x 2.45 x 3 m with a trailer of 8m x 2.45 x 3m to make sure it can hold orders of 100 and more orders.
- Aerothene Foam wrap is 1,5m x 200m of wrapping on one roll. The dimensions of one roll are therefore 1,5 x 1 m
- Assume 1 Liter of paint covers approximately 10 m^2.
- Assume the plastic handlebar size as 30 mm diameter, 125 mm long with a weight of 0.15kg per piece.
- Assume a timber pallet will be used for steel sheets with dimensions 3m x 1.2m that weighs 47 kg and can handle heavy duty loads, of max 1500 kg.
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6.Needs Analysis

6.1 Product design

In the product design we will establish the products to be produced and the specifications of the individual parts.

6.1.1 Raw material unit

Table 1:Raw material unit

Type of material	Size
Steel sheets	2,5 x 1,2m x 4,8 mm thick
Steel tubes	3,6 m(L) x 30mm thick
Plastic pellets	Bought in 25kg bags
oil	Stored in a 2500L tank
Wheels	33 cm (L) x 10cm (w)
Paint	0.55L
Boxes	150 x 75 x 70
Bolts	50mm (L) x 10mm (diameter)
Washers	20mm diameter
Nuts	15mm (W) x 10mm (H)
Aerothene foam wrapping	200m x 1.5 m

To calculate the weight of the steel sheets and steel tubes we will assume the density of steel as 7850 kg/m^3.

Steel sheet:

$$Volume = length * \ Width * \ Thickness$$

$$= 2.5\ m * \ 1.2m * \ 0.0048m$$

$$= 0.0144\ m^3$$

$$Weight = 0.0144m^3 * 7850 \text{ kg}/m^3$$

= 113.04 kg p.steel sheet.

Steel tubes:

$$Volume = \ \pi * (\frac{0.03}{2})^2 * 3.6$$

$$= 2.54 * 10^{-3}$$

$$Weight = 2.54 * 10^{-3} * 7850\frac{kg}{m^3}$$

$$= 19.95 \ \approx 20kg\ p.steel\ tube.$$

6.1.2 Final product unit

Table 2:Final product unit

Final product	Size
Bucket	80 x 60 x 20cm
Frame	120 x 40 x 60 cm
Handlebars	125 (L) x 30mm (thick) x 0.15kg
Final wheelbarrow	141 x 67 x 60 cm x 5.4 kg

6.1.3 Lot and order size

Large orders are orders bigger than 100 where we will use our own fleet and small orders are less than 100.

6.2 Process design

In the process design we will specify the required manufacturing processes and related activities needed to produce the final products.

6.2.1 Process flow diagram

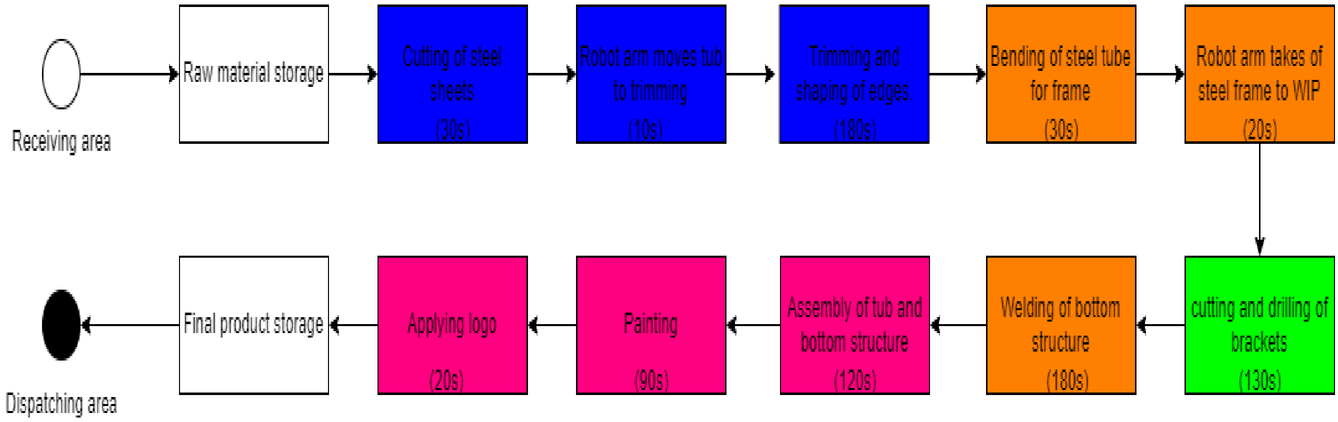

Figure 1: Process flow diagram

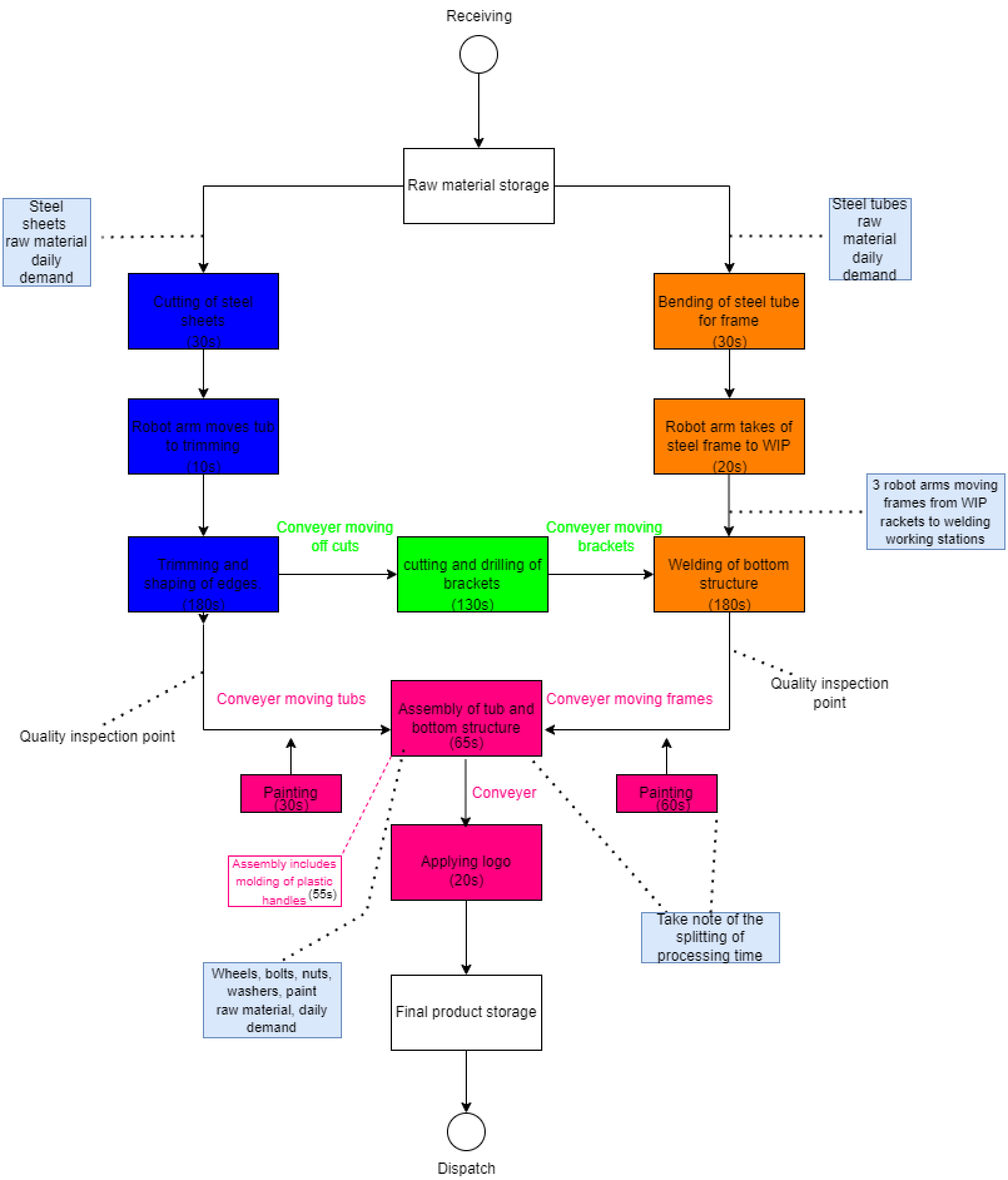

Figure 2:Detailed process flow diagram

6.2.2 Required Processes

Cutting of Steel sheets

Figure 3:Hydraulic press machine

Harsle Four column Hydraulic press Y27-400T used for shaping.

This is the first step of the shaping of the body of the wheelbarrow. After assembling the drawing die, the metal sheet is drawn into the basic shape of the body to make the shape of the tub.

Dimensions: 2600 mm x 5800 mm

Robot arms

Figure 4:Robot arm

The Borunte 6 Axis Robot Industrial Robotic arm which has a max of 1500 mm arm length and length load of 10 kg will be used to move the tub from the first hydraulic press to the second hydraulic press for the trimming. It will also be used to move the wheel bend frame from the bending machine to the WIP station.

Trimming

Figure 5:Hydraulic trimming machine

Harsle 4 column hydraulic press Y32-160T.

Now after installing the die, the shaped tube will be placed to trim the edge of the semi-finished tube that has been drawn in the previous step.

Dimensions: 2400 x 1300 x 3200 mm

Crimping

Another Harsle 4 column hydraulic press Y32-160T will be used to shape the edges more firmly and round.

Dimensions: 2400 x 1300 x 3200 mm

Punching

The Harsle 4 column hydraulic press Y32-100T (smaller tonnage) can be used to punch the 6 holes in the bucket. The positions of the holes must be determined by placing the punching die to prevent a defect and then it can be replaced by the bucket for the punching.

Dimensions: 2200 x 1200 x 2800 mm.

Bending of steel tubes

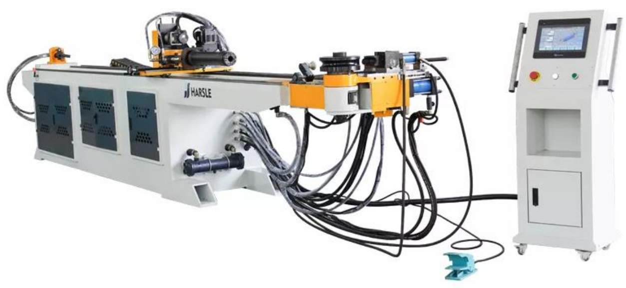

Figure 6:Bending machine

Harsle Tube bending machine 38CNC-2A-1S.

This machine will be used to bend the steel tubes to shape the frame of the handlebar, legs and wheel bend for the wheelbarrow.

Dimensions: 5500 x 1300 x 1200 mm.

Rackets (WIP)

Brackets

Figure 7:Milling machine

Haas VF-2 Milling Machine.

Dimensions: 2500 mm x 2000 mm x 2500 mm

The milling machine will be used to shape all the brackets needed by using the off cuts from the steel sheets.

Welding

Figure 8:Welding machine

Megatic MIG 205 Welding machine helps to use the space efficiently as it is not a big machine and would mostly require working space for the operator using it. Its specifications align well with the requirements for welding mild steel components including the brackets to the wheelbarrow frame. To obtain the best welding quality, it is important to make sure the specified welding parameters including voltage, current, wire feed speed are properly set for the thickness and kind of material being welded.

Plastic caps

Figure 9:Injection molding machine

Engel Victory 200T Injection molding machine.

The injection moulding machine will be used to produce the plastic handle caps.

The plastic pellets will be loaded into the hopper, then it will automatically melt, the molten cavity of the desired handle cap will be injected, and the final plastic handle caps will be ejected after the plastic cooled down.

Dimensions: 6000 x 1900 x 3100 mm

Painting and stickers

Figure 10:Painting gun

Bash PFS 3000-2 Paint spray system with a 650 W motor which makes it quite time efficient and it has a 1Lpaint container and a 3m pipe which gives the operator space to do the painting in a comfortable manner. These painting guns will be used to paint the wheelbarrow frames and buckets separately as well as the logo with a stencil.

Packaging

Figure 11Wrapping machine

The final products boxes on the pallets will be wrapped with the automatic packaging pallet wrapping machine 2M_XT45051. The dimensions of the machine are 6000 x 1650 x 3080 mm. It will use the aerothene foam wrapping.

6.3 Schedule design

The schedule design will determine the quantitative values of the products to be produced.

6.3.1 Production requirements

Assuming we have 5 working days in a week, and 20 days in a month, means 240 working days a year. The final product demand is as follow:

Table 3:Production requirements

Product	Demand p/day	Demand p/week	Demand p/year
Wheelbarrow	1000	5000	240 000
Frames	1000	5000	240 000
Handle caps	2000	10 000	480 000

6.3.2 Production Capacity

The production capacity will determine the number of machines needed to meet production requirements. It will be calculating by using the processing times for each machine to determine how much machines are needed at a certain working station.

$$Production\ time = 8hour\ shift * 60 = 480\ min = 28800 \text{ seconds.}$$

$$Total\ cycle\ time = 30 + 10 + 180 + 30 + 20 + 130 + 180 + 120 + 90 + 20$$

$$= 13.5 \min for\ one\ unit.$$

Note that this total cycle time does not include the 2 quality inspection stations that is after the bucket shaping process and the welding of the bottom structure which could possibly cause bottlenecks.

$$Required\ output\ rate\ p/\min = \frac{1000\ wheelbarrows}{480min}$$

$$= 2.0833\ wheelbarrows\ p/min$$

Note that the trimming and shaping of tub edges are split into 3 workstations with different machines, therefore the total machines are 9 for this part of the process. The Assembly of the tub and bottom structure includes 2 separate workstations, first the moulding of the plastic handle caps and secondly the assembly of the tub with the bottom structure with the wheel and applying the handle caps. Therefore, this process time given of 120 s are split into 55 seconds for the moulding of plastic caps and 65 seconds for the second workstation of the assembly process. The painting process is also split into 2 different workstations, first the painting of the bucket and then the frame, therefore the 90 seconds given is split into 60 seconds and 30 seconds for the 2 different workstations. The applying of the logo will not be in the form of stickers but rather be painted with a stencil and another painting gun, for more efficiency.

Table 4:Cycle times of machines

Process	**Time**	**Units**	**Machines**
Cutting of steel sheets	28 800/ 30 s	960	2
Moving tub to next process (robot arm)	28 800/10 s	2880	1
Trimming and shaping of tub edges	28 800/60 s	480	3x3 = 9
Bending of steel tube	28 800/30 s	960	2
Robot takes frame to WIP	28 800/20 s	1440	1
Cutting and drilling of brackets	28 800/130	220(set)	5
Welding of bottom structure	28 800/ 180	160	7
Painting of bottom structure	28 800/30	960	2
Moulding of plastic caps	28 800/55	523	2
Assembly of tub and bottom structure	28 800/65	443	3
Painting of tub	28 800/60	480	3
Applying logo	28 800/20	1440	1

Now that the cycle times for each workstation is calculated the exact amount of each machine can be determined:

Table 5:Amount of machines required

Machine	Amount	Operators
Hydraulic press Y27-400T	2	2
Hydraulic press Y27-160T (x2)	6	6
Hydraulic press Y 32 - 100T	3	3
Tube bending machine	2	2
Milling machine	5	5
Welding machine	7	7
Injection Moulding machine	2	2

Painting guns	6	6
Robot arms	6	0

6.4 Raw material requirements

To produce the number of different products required for the final wheelbarrow, different raw materials will be used and the amount that is needed is calculated below.

We will assume an estimate of 250ml oil is used per metal sheet for both sides.

Calculating the surface area of the wheelbarrow to determine the amount of paint needed:

$$Inside/outside\ surface\ area:$$

$$Bottom: 80cm * 60cm = 4800\ cm^2$$

$$2\ long\ sides = 2(80 * 20cm) = 3200\ cm^2$$

$$2\ short\ sides = 2(60 * 20cm) = 2400cm^2$$

$$Total\ inside/outside\ area = 4800 + 3200 + 2400 = 10400\ cm^2$$

$$Outer\ rim\ area\ (assuming\ 2cm\ rim\ width):$$

$$Rim\ length = 2(80 * 60cm) = \ 280\ cm$$

$$Rim\ area = 280\ cm * 2cm = 560\ cm^2$$

$$Total\ area = 10400(2) + 560 = 21\ 360\ cm^2 = 2136\ m^2$$

Assuming 1L paint covers 10 m^2.

Paint required for tub:

$$= \frac{surface\ area}{paint\ coverage\ p.L}$$

$$\frac{2136}{10\ m^2.liter} = 0.2136\ L\ required\ p.tub(green)$$

Paint required for frame:

$$Cylinder\ surface\ area$$

$$A = \ \pi * d * L$$

$$= \pi * \ 0.03 * 3.6$$

$$= 0.3393\ m^2$$

$$Paint = \frac{0.3393}{10\ m^2.liter}$$

$$= 0.03393\ \approx 0.034\ L\ required\ per.frame\ taking\ brackets\ and\ logo\ into\ account.$$

To calculate the amount of wrapping plastic that is needed ton wrap the final products which is a pallet with 8 boxes (3m x 1.2m x 1,55 m) we will calculate the surface area:

$$Front\ and\ back\ area: 2 * (3 * 1.55) = 9.3m^2$$

$$Left\ and\ right\ area: 2 * (1.2 * 1.55) = 3.72m^2$$

$$Top\ and\ bottom: 2 * (3 * 1.2) = 7.2\ m^2$$

$$Total\ surface\ area = 9.3 + 3.72 + 7.2 = 20.22\ m^2$$

For the daily requirement of wrapping, we will need 22 200 m^2 of wrapping. One unit equals a roll of 1.5 m x 200m. Therefore, we will need 74 rolls of the wrapping.

It can be seen as a bill of material that indicates the amount of raw material needed to produce one wheelbarrow and the amount per day to meet the daily demand.

Table 6:Bill of materials

Raw material	Amount p/day	Amount p/wheelbarrow
Steel sheets	1000	1
Steel tubes	1000	1
Oil	250L	250ml
Brackets	6000	6000ml
Plastic Pallets	300 kg	0.3 kg
Bolts	8000	8
Washers	6000	6
Nuts	8000	8
Wheels	1000	1
Paint for tub	210 L	0.21 L
Paint for frame	34 L	0.03393 L
Wrapping	22 220m^2	22.22 m^2
Box	1000	1

6.5 Flow systems

The flow systems will describe how the materials will flow through the different departments as well as how it will be stored to meet the raw material, final product and WIP inventory demands.

6.5.1 Material management system

The material management system considers the flow of the material into the system, mostly the raw materials inventory. The different raw materials will be arriving via trucks from our suppliers. The trucks will park at the arrival area and the products will be unloaded using forklifts.

Figure 12:Steel sheets packaging

The steel sheets will arrive on timber pallets (3m x 1.2m) which is a bigger pallet than the standard size to accommodate the dimensions of the steel sheets (2.5m x 1.2m). The timber pallets can hold heavy loads and therefore 10 steel sheets will be packed on a pallet which gives a total weight of 1130 kg for the 10 sheets as the weight of the steel sheets are previously calculated as 113kg. This will give a height of 0.048 m + the pallet height of 0.15m and a additional height of the wrapping 0.002m equals a total height of 0.2m. A few of these pallets can therefore be stacked on top of each other in the storage area, to use space efficiently, as the height is not so big.

Figure 13:Timber pallet

The pallets will be stacked 10 levels high on the static pallet racks shown below as one pallet with 10 steel sheets will weigh approximately 1000 kg and can therefore not be stacked on each other.

Figure 14:Static pallet racks

The steel tubes will arrive in this standard wrapping of 75 tubes together.

Figure 15:Steel tubes bundles

Figure 16:Steel tubes racks

Figure 17:Steel tubes racks doublesided

The bundles of 75 stubes will be packed on cantilever racking which is very convenient as you can adjust the arms where you want to and decide how much levels high, how long each rack will be and if it is going to be double sided or put up against a wall. These racks are effective as it saves a lot of space, and it can hold up heavy loads. We will therefore customise it with 3 arms per set of a rack that is 4 m long and make it 5 levels per rack. The bundle of 75 tubes will equal a mass of 1600 kg as the weight of a steel tube was previously calculated as 20kg.

Figure 18:Plastic bin packaging

We will receive the wheels packed in a plastic bin (1200 x 1000 x 750mm) which can take up to 1000 kg. The wheels will be packed in with its diameter showing to the width of the bin and the width of the wheel (10cm) on the length side of the bin, therefore there can fit 36 x 100 wheels in one level of the bin, and we will pack it 2 levels high, so there will fit a total of 720 wheels in one bin. These bins are closed with a lid and can therefore be stacked on each other. It is the exact size of a standard pallet, so there will be 2 bins on one pallet with a height of 1650 mm including the pallet height.

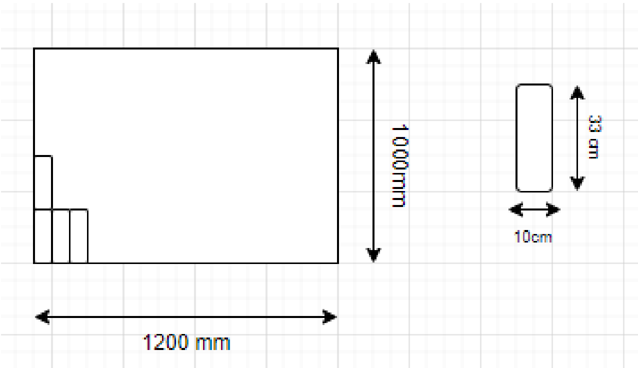

Figure 19:Wheel packaging alignment

The bolts, washers and nuts will be received in the same type of plastic bins, each separate in a unique size of bin stacked on a standard pallet. We will store it just like we received it. The bolts will be stored in the same size (1200 x 1000 x 750mm) bin. There will be 2 bins stacked on a standard pallet. The volume of the bin can be calculated as:

$$Volume = L * W * H$$

$$Volume = 1200 * 1000 * 750mm = 900\,000\,000\; mm^3$$

Volume of bolt (assuming it as a cylinder): $\pi * 5^2 * 50 = 3926\; mm^3$

Number of bolts in bin: $\frac{900\,000\,000mm^3}{3926mm^3}$ = 229 361 bolts.

For the nuts, the same type of bin will be used but with the size of 594 x 338 x 280 mm(outside) with inner dimensions of (510 x 338 x 280 mm). The volume of the bin can be calculated as:

$$volume = 510 * 338 * 280\; mm = 47\,026\,400\; mm^3$$

Assuming the size of a nut like a rectangular prism:

$$Volume\ of\ nut = 15mm * 10mm * 10mm = 1500mm^3$$

$Number\ of\ nuts$: $\frac{47\,026\,400\; mm^3}{1500\; mm^3}$ = 31 351 nuts.

For the washers, we will use the same type of plastic bin but with a size of 400 x 297 x 315 mm (outer dimensions) and inner dimension 342 x 235 x 280 mm.

$$Volume\ of\ bin = 342 * 235 * 280\; mm = 22700400\; mm^3$$

Assuming size of washer as a circle:

$$Volume\ of\ washer: \mu * 10^2 * 2 = 628\; mm^3$$

$$Number\ of\ washers\ in\ bin = \frac{22\,700\,400}{628} = 36124\; washers.$$

Figure 20;Oil and paint storage

We will store the amount of oil needed for raw storage inventory in a Rototank. The oil will not degrade in the plastic tank as it will stand for a maximum of 2 weeks in the pump. The size needed for the oil tank will be 2500 L tank with dimensions 1400 mm(D) x 1750 mm (H). We will receive the oil with an oil supply truck which will pump it into our tank.

The paint will also be stored in similar tanks but a smaller size, as we don't need as much paint as oil. We will store the black paint for the frames and logo and green paint for the buckets in separate tanks. The tank for the green paint will be 2200 L with dimensions 1300 mm(D) x 1800 mm (H). 2 plastic drums 210L with dimensions 58cm x 102cm will be used.

Figure 21:Plastic Pellets

Figure 22:Plastic Pellets packaging

The plastic pellets that will be used for the plastic handle caps, will be bought in 25kg bags that are stacked on standard pallets. The sizes of the bags can be assumed as 60 x 40 x 15 cm. 4 of these bags are packed on the pallet and will be stacked 10 levels high and wrapped with plastic as seen above, therefore a total of 40 bags on one pallet.

Figure 23:Final product packaging

The boxes for the final product packaging are also part of the raw material required. It will be bought in stacks packs of 200 which is therefore 2m high per stack, therefore 5 packs will be needed for 2 weeks inventory.

The foam wrapping rolls will just be stacked one layer on each other. 1 week of inventory will be stored in the raw material storage and the daily demand will be stacked at the wrapping machine in the final product storage area.

6.5.2 Material flow systems

The material flow system is how material and products will flow within the facility and mostly define what is part of the Work-in-Progress (WIP).

100 pallets of the steel sheets will be taken to the hydraulic press to meet the daily demand needed.

There will be 3 of the same racks used in raw material storage against the wall used for the steel tubes but only one sided to use space efficient, for the daily demand of 14 bundles of steel tubes. These racks will be put up against the wall where the bending machine are located. It will take up 12 x 1 m space.

4 bins of wheels will be taken to the assembly workstation from the raw material storage to meet the daily demand needed. 12 bags of the plastic pellet bags will be taken to the injection moulding machine daily to meet the demand.

The frames of the wheelbarrows will be hanged on the rackets that are part of the WIP inventory. The rackets can be customised and therefore we will make one racket a double level of 50 x 2 x 3m, assuming 250 frames would fit on one level and therefore we will need 2 of them for the demand.

Figure 24:WIP frames

The off cuts of the steel sheets when trimming the tubs will be moved into a big container, which is also WIP because it will then be used for the making of the brackets by the milling machine.

There will be a similar tank that is used in the storage area for the oil and paint but a smaller size that can accommodate approximately 2 days demand for the oil and paint which will be standing at the workstations that it is needed. These tanks will have to be located near the storage tanks so that it is easy to pump it from the one tank to the other tank. It is calculated that one steel sheet needs 250 ml oil, therefore 2500 L of oil is needed per day and a 500 L tank will be used with dimensions 840(d) x 1080 mm (h) that will stand right beside the Hydraulic press machine. A daily demand of 210 L green paint is needed for the tubs and will be stored in a 210 L drum that will stand at the painting workstations. 34L black paint is needed per day for the frames and will be stored in 2 x 20L paint buckets. The black paint will be poured daily in these buckets from the big tank and carried to the workstations for the daily demand.

One bin for the bolts, washers and nuts will be moved to the assembly workstation from the raw material storage and will be filled up as needed.

2 of the static pallet racks will be put up for 5 levels high to stock the 10 pallets of daily demand in the bucket department.

6.5.3 Physical distribution system

The physical distribution system includes the way in which our final products will exit the facility. Each wheelbarrow will be packed into a box with dimensions 150 x 75 x 70 cm. The timber pallets (3m x 1.2m) will be used and 4 boxes can be packed in one level of the pallet. Each pallet will include 2 levels, therefore 8 wheelbarrows on one pallet that is wrapped in aerothene foam wrapping for transporting. 5 Operators will be used to pack the wheelbarrows in the boxes.

6.6 Material handling equipment

We will be using an overhead crane at the raw material storage as well as at the final product storage. This effective as it does not take up any floor space as it is put up high above everything.

Forklifts will be used to unpack the raw materials from the delivery trucks, moving most of the raw materials in the manufacturing system to the needed workstations and for the final products into the dispatching trucks. The standard size of a forklift is 3800 x 1230 mm, and one operator is needed for each forklift. We will be using fork extensions as the standard lengths ranges from 1.2 m to 2.4 m, so that it would be able to move the 3.6 m steel sheets and tender pallets which is 3m long. It would be 4-wheel forklifts so that it can move weights of up to 3600kg.

To calculate the number of forklifts needed we will assume theoretically that all the raw materials will be delivered on the same day to determine the max volume it should handle. The total number of pallets of raw materials include 1146 pallets. This includes a bundle of steel tubes as one pallet.

$$cycle\ time: 10\ \mathrm{min}\ per\ pallet\ move\ (unloading, transport, loading)$$

Forklift capacity per hour:

$$= \frac{60\ min}{10\ min} = 6 \text{ pallets}$$

$$Total\ pallets\ per\ h = \frac{1146}{8\ h} = 144 \text{ pallets.}$$

$$Forklifts\ needed = \frac{144}{6} = 24 \text{ forklifts for raw materials.}$$

Assume the whole 1-week inventory of final products will also be dispatched to calculate the max number of forklifts needed in the dispatching area. This is a total of 625 pallets. Doing the same calculations equals 13 forklifts needed. Taking 90 % utilization rate of the forklifts into account, determines we need 2 extra forklifts for the raw material storage and 1 extra forklifts for the final product storage. This equals a total of 40 forklifts. Assume these forklifts will also be used as needed to move to and from the manufacturing parts, as this was calculated as the max number of forklifts needed.

A robot arm will be used to move the tubs from the shaping workstation to the trimming workstation as well as at the tube bending machine to move the frame to the WIP rackets.

A conveyer belt system will be used where the buckets will be hanged after it is shaped and trimmed and then it will move past the painting operators to the assembly workstation. Another system will also be used where the welded frames will be hanged to go past the painting operators to the final assembly station. This is a very effective system as the painting can be done while it is transported from the one workstation to the other. The length of the conveyer belt will be customised as the length between the workstations.

6.7 Storage

In the following table the calculated number of units will be shown that will be stored to meet the demand of raw material storage of 2 weeks. These units are defined previously in the flow systems how all the different materials are stored in these units. Note that the demanded steel tubes are 134 bundles of 75 steel tubes each. Therefore, we will need 14 racks that we will be double sided with 5 levels each to store these bundles on. The racks are 4m in length and 2 m (double sided) width, therefore the total storage it will take is 56m long x 2 m wide.

Table 7:Material inventory required

Material	Amount
steel sheets	1000 pallets
steel tubes	134 bundles
wheels	14 bins = 7 pallets
Bolts	4 bins = 2 pallets
Washers	2 bins = 0.5 pallets
Nuts	2 bins = 0.5 pallet

Oil	1 x 2500L tank
Black paint	2 x 210 L plastic drums
Green paint	2200 L tank
plastic pellets	3 pallets = 120 bags

Note that the final products boxes are stacked in pairs of 4 boxes per level, using the timber pallets again (3m x 1,2 m) stacking it 2 levels high that equals 8 boxes on one pallet with a total height of 1550 mm including the height of the pallet. We will therefore need 625 pallets in the final product storage to satisfy the need of 1 week demanded products.

6.8 Personnel requirements

6.8.1 Operators

We will assume 1 operator for each machine required at the working stations as well as the material handling equipment.

Table 8:Amount of operators per machine

Machine	**Operators**
Hydraulic press Y27-400T	2
Hydraulic press Y27-160T (x2)	6
Hydraulic press Y 32 - 100T	3
Tube bending machine	2
Milling machine	5
Welding machine	7
Injection Molding machine	2
Painting guns	7
Total	34

Table 9:Operators per material handling machine

Machine	Operators
Forklifts	40
Overhead cranes	2
plastic wrapper	1
Truck drivers	5
Total	48

6.8.2 Management and office staff

The following management staff will be needed to run the various departments of WB equipment. They will be located in the office departments and will be able to move to any of the other departments. The table also includes total operators.

Table 10:Management staff

Staff	Amount
Management	3
Administrative	5
Supervisors	4
Operators	80
Total	92

6.8.3 Service staff

The service staff will perform service jobs to assist the facilities main operations.

Table 11:Service staff

Staff	**Amount**
Maintenance	2
Cleaners	3
Security	2
Receptionist	1
Total	8

This gives us a total of 184 staff members.

6.9 Space requirements

In the space requirements we will determine the total amount of space needed for our facility. We will split the facility into departments but also the departments into workstations.

6.9.1Raw material Storage

Table 12:Raw material storage

Product	Calculation	Size(m^2)
Forklifts parking	26(3.8 x 1.23m)	122
Steel sheets	(6 x 5) x (3,6 x 4)	432
Forklifts moving space	(4 x 3,6 x 5) + (4 x 50) + 12	284
Steel tubes	12 x 56	672
Wheels,Bolts, Washers, Nuts, Plastic pellets	4+	17
Tanks	6 x 2	12

Carton boxes	14.5 x 2.15	31
Total		1570

6.9.2Final product Storage

Table 13:Final product storage

Product	Calculation	Size(m^2)
Pallets	6 x 3,6 x 5 x 6	648
Boxes	1 pack = 2.9 x 2.15	6
Wrapping machine	6 x 1.65 x 3	30
Wrapping rolls	19 x19	361
6 operators	6(2)	24
Forklifts parking	14(3.8 x 1.23)	65
Forklifts moving between pallets	(4 x 3.6 x 5 x 6) + (50 x 4 x 6)	1632
Total		2766

6.9.3Bucket department

We will give each operator (2 x 2m) space.

Table 14:Bucket department

Machine	Nr	Dimensions	Total machine space	Operators	Material space	Total
Hydraulic press Y27-400T	2	2,6 x 5,8 m	30	2 x (2 x 2)= 8 m^2	2 x 3 x 2=12	$50m^2$
Hydraulic press Y32-160T	6	2,4 x 1,3 m	19	6X(2x2) = $24m^2$		$43m^2$
Hydraulic pressY32-100T	3	2,2 x 1,2 m	8	3x(2x2) = $12m^2$		$20m^2$
Robot arm	1	4 x 4m	16			$16m^2$
Total						$129m^2$

6.9.4Brackets department

Table 15:Brackets department

Machine	Nr	Dimensions	Total machine space	Operators	Material space	Total
Milling machine	5	2.5 x 2m	25	5 x (2x2) = 20		45
Conveyer belt	2	(14 m long x 1m)	28			28
Total						73

Note that there will no material space be needed, as the cut offs that are used as the raw material will move directly on a conveyer belt from the trimming machine in the bucket department to the milling machines. The WIP for the final brackets is also the second conveyer which will automatically move the brackets to the welding workstation from all the milling machines.

6.9.5Frame department

Table 16:Frame department

Machine	Nr	Dimensions	Total machine space	Operators	Material space	Total
Bending machine	2	5.5 x 1.3m	14.3	2 x (2 x 2) = 8	3 x 1 x 4 = 12	34
Forklifts		12 x 4m	148			148
WIP rackets	2	50 x 2 m	200			200
Welding	7			7 x (4 x 4) = 112	3 x3 =9	72
Robot arms	5	4 x 4	80			80
Total						534

Note the frame department includes the making of the frames with the racket for the daily needed steel tubes and the forklift space needed to deliver it, the WIP rackets and the robot arm that hangs the frames on the rackets, and then the welders whose operating space includes them doing the job and having a small bucket with brackets. There will be a big container (3 x 3m) where the brackets will be dropped in from the conveyer and then each welder will have to refill their small buckets with the brackets. There will be another 2 robot arms that will give the frames to all the welders automatically. The working space for the welding operators are bigger than the standard working space for other machine operators, as this welding working space includes the welding machine, the frame that they are welding and their own bucket with brackets.

6.9.6Assembly department

There will be a conveyer belt moving the assembled frames from the frame department and another conveyer belt that will be moving the buckets from the bucket department to this apartment. There will be painters assigned that will paint the frames and buckets with the separate green and black paint while it is moving past them on the conveyer belt. Then after it is painted there will be a robot arm that moves the frame and bucket to each assembler's working station. Note that the assembly material space includes one pallet with a containing the wheels, 2 pallets including the bolts, nuts, and washers, one (2 x 2m) bucket containing the plastic handle caps ((1 x 1.2) + (2 x 2) + 2(1.2 x 1)) which equals a total of $8m^2$ material space. After the wheelbarrow is fully assembled it will be hanged on another conveyer belt which will move it to the final product storage. While it moves there will be another painter painting the logos on the wheelbarrows.

Table 17:Assembly department

Machine	Nr	Dimensions(m^2)	Total machine space(m^2)	Operators	Material space	Total(m^2)
Injection molding machine	2	6 x 1.9 m	23	2 x (2 x 2)= 8	1.2 x 1 = 1.2	32
conveyer belts	3	5 x 2	30			30
Frame painting	3			3 (3 x 3) = 27	included	27
Bucket painting	3			3 x (3 x 3)	included	27
assembly	3			3 x (3x3) = 27	8	35
Robot arm	2	4 x 4	32			32
Logo painter	1			3 x3 = 9		9
Total						192

6.9.7Receiving department

Distributing 10 x 4m space for each truck so a forklift can move around it and distributing space for max 5 trucks at once equals 200 m^2. The unloading bay includes a space of 9 m^2 $for\ each\ truck$. It will also include one bathroom of 10 m^2Total area = 255 m^2.

6.9.8Dispatch department

Distributing 10 x 4m space for each truck so a forklift can move around it and distributing space for max 5 trucks at once equals 200 m^2. The loading bay includes a space of 9 m^2 per truck which equates to a total of 90 m^2 including our own fleet. It will also include one bathroom of 10 m^2 for the truck drivers.

The dispatch area will also include the parking of our own fleet giving 15 x 4 m for each truck, having 5 trucks equals 300 m^2 which gives a total of 590 m^2.

6.9.9Personnel service requirements

A standard rule for open office space per person is 4.5 m^2per desk and 4.6 m^2per person for communal areas. This equals to a total office space of 9.2 m^2per staff member. We will assume half of our personnel makes use of public transport and the other half has their own cars + 5 extra parings for visitors. A standard parking space will be 6.7 x 2.6 m with an aisle space of 7 m width. Our cafeteria will include a serving line where workers will receive their food and eat it at the dining area. The cafeteria will include a kitchen area and seating area. The industrial setting space requirement for a cafeteria is 1.4 m^2 per person. The total seating area equals 1.4 x 178 = 250 m^2 The kitchen area will be 100 m^2 to serve between 180 people.

According to the Health and Safety Act of South Africa for every 20 employees, there should be 1 bathroom. We will also account for 5 extra truck drivers' bathrooms at the receiving and dispatch area. A bathroom includes a toilet, wash basin and a locker on the outside wall next to the door. The standard size of a bathroom is 9.3 m^2. The reception area will include space for clients, visitors and stakeholders to enter the facility. It will include space for one receptionist with a desk to receive visitors and a waiting area. We assume it can accommodate 6 visitors at once. The reception will also include its own bathroom.

Table 18:Personnel requirements

Area	Calculation	Total space(m^2)
Office	9.2 x 12	111
Parking	6.7 x 2.6 x (89+5)+(47 x 2.6 x 7)	2493
Cafeteria	250 +100	350
Bathrooms	9 x 9.3	84
Reception	4.5 + 4.6(7) + 9.3	46
Total		3084

6.9.10 Total space requirements

We will distribute a 20 % aisle space of the total space requirements as well as an extra 20 % aisle space for the forklifts. There is already forklift moving space calculated and included in the main areas, so this extra 20% is for extra safety and to make sure it can move through the whole plant.

Table 19:Total space requirements

Area	Space (m^2)
Raw material storage	1570
Final product storage	2766
Manufacturing departments	860
Personnel facilities	3084
Aisle space	1656
Forklift space	1656
Total	11592

7.Design Alternatives

7.1 Activity relationships

The relationships between the departments will be analysed to determine the arrangement of the facility.

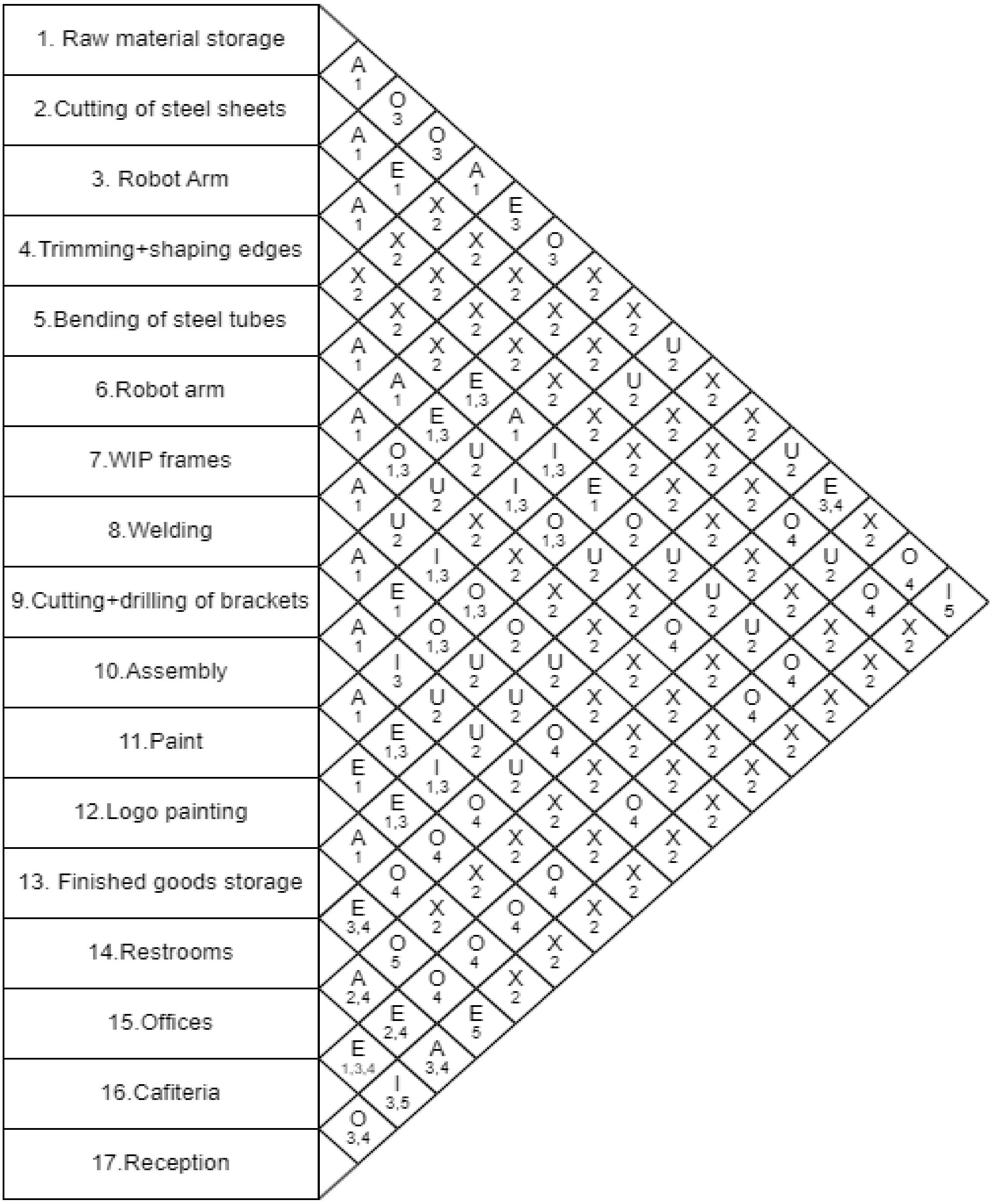

Figure 25:Relationship Chart

Table 20:Chart key

Closeness	Value

Absolutely necessary	A
Especially important	E
Important	I
Ordinary Closeness OK	O
Unimportant	U
Not desirable	X

Table 21:Chart reasons

Reason	Code
High frequency of flow	1
Low frequency of flow	2
Minimize distance travelled	3
Convenience flow	4
Information flow	5

The workstations that are important to be together (as grouped by department) are graded as important to promote flow and decrease the distance of the WIP to travel. This will promote efficiency as each part can move from the one operation to the next. It is important for the reception to be close to the receiving and dispatch departments as it is necessary for effective coordination of incoming or outgoing shipments and smooth flow of information between them. Note that the raw material storage area in the chart can be seen as the receiving area and the finished products storage can be seen as the dispatching area in the chart as it is assumed that these 2 will be together.

7.2 Designs

The following block layouts represent possible layouts for our design problem as solutions. The activity relationship chart was used to determine which departments are important to be close to each other for various reasons and will be taken into consideration when designing the layouts. There is a colour convention table where each department has a different colour to simplify the design. The legend shows other symbols that are implemented. Note that the designs are drawn on exact scale where each small block in the background represents 4 x4 m, as there are 4 x 4 small pixels in each block that cannot be seen. The dimensions are shown around the designs in meters. Note that in each design, there are restrooms included in the receiving and dispatch areas for the truck drivers or workers in that section, as the restrooms are most of the time far away from these departments. This will help to reduce potential bottlenecks and delays in the process.

Table 22:Alternatives colour key

Offices	
Parking	
Restrooms	
Reception	
Cafeteria	
Receiving	
Dispatch	
Security	
Assembly department	
Brackets department	
Frame department	
Bucket department	
Finished goods storage	
Raw material storage	

Table 23:Alternatives legend

Legend	
	Operational flow
	Gate
	Trucks flow
	Begin and end of operational flow

Alternative 1

In this alternative there are two gates. The trucks and staff will both enter at the same gate where they can move to their separate directions. The second gate is only for the exiting of the trucks as they will follow a linear flow, but the staff will exit at the main gate 1 as it is located at their parking so there is no need to go to the other gate. These separate exit gates helps reducing risks of traffic congestion.

The raw material and final products storage area are beside each other where the operational flow begins. The operational departments are placed in a U-shape where they begin and end at the storage areas while the supporting facilities are are grouped together in the middle of the U-shape. The staff and visitors parking area are above the U-shape, to make sure the staff can still enter the facility at the reception without having to intersect the operational area.

A big problem with this layout is that the bucket department is out of reach of the assembly area, as it was determined in the space relationship chart that it is absolutely important for these departments to be close to each other. The operational flow follows a linear line, where all the departments follow directly on each other, but it is not effective for this type of operation as it is necessary for all the operational departments to be close to each other as the workstations crosses each other and promotes minimised distance travelled of the WIP materials. Another problem is that the reception blocks the flow between the raw material storage and the bucket department.

The total space the design acquires is 9243 m^2 which provides for more than 30 % of potential future expansion.

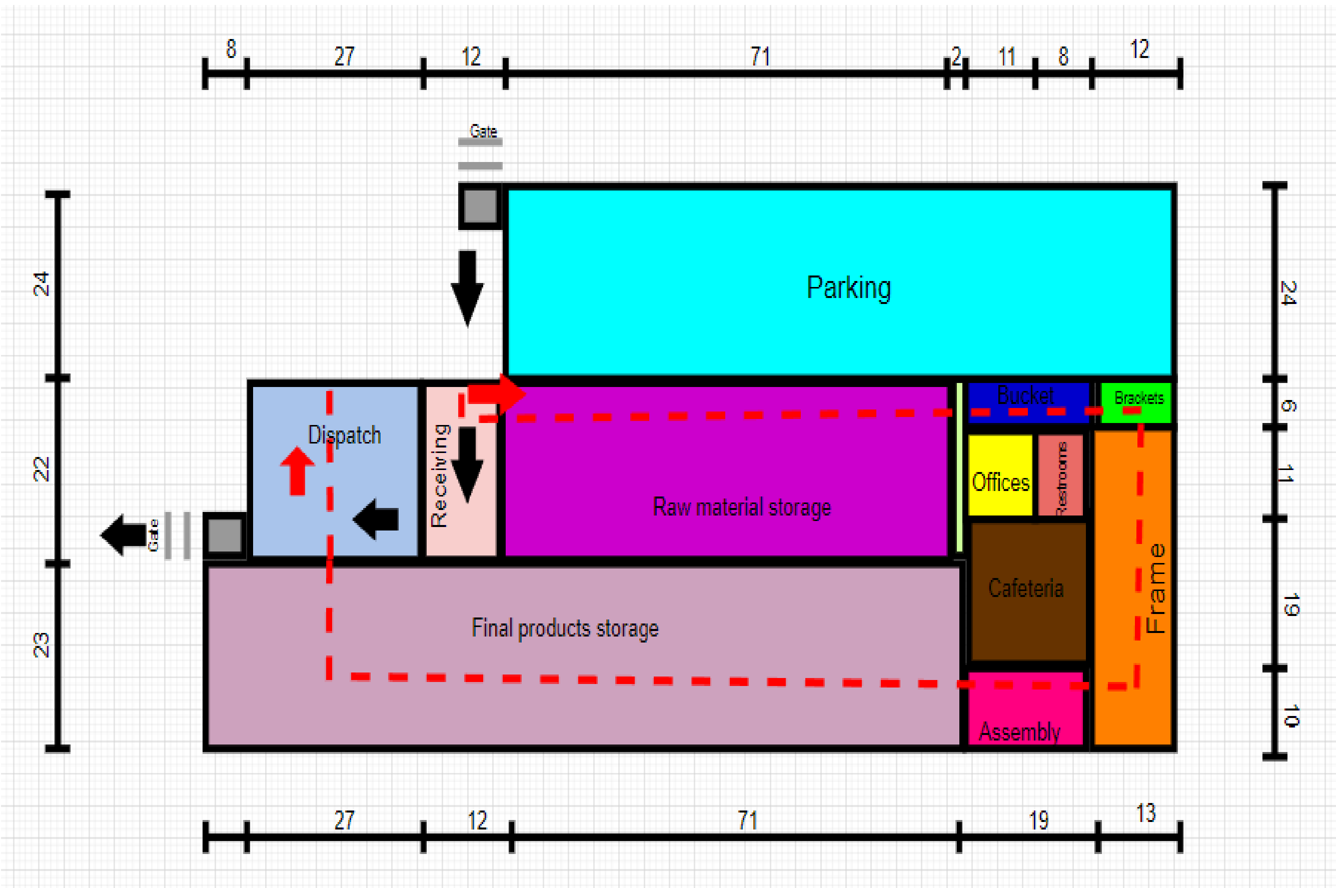

Figure 26:Alternative 1

Alternative 2

In this alternative there are 3 gates, one for the staff and visitors entering the parking and the other 2 is on the other side of the facility for entrance and exit of trucks. This reduces the risk of possible traffic congestion as they will not pass each other.

The receiving and dispatch areas are beside each other as well as the storage areas which also promotes flow and minimises distance travelled, where all the trucks can enter at the one gate, on a linear path and exit again after they have received or delivered.

The operations are grouped together in a combination of importance and the type of operation, while the storage area is shaped as a U-shape around the operations, with the other supporting facilities on the side that closes the U-shape. This layout promotes operational efficiency as it is designed in a way to not waste any area space.

The total area space the design acquires is 9194 m^2. This provides space for more than 30% potential future expansion.

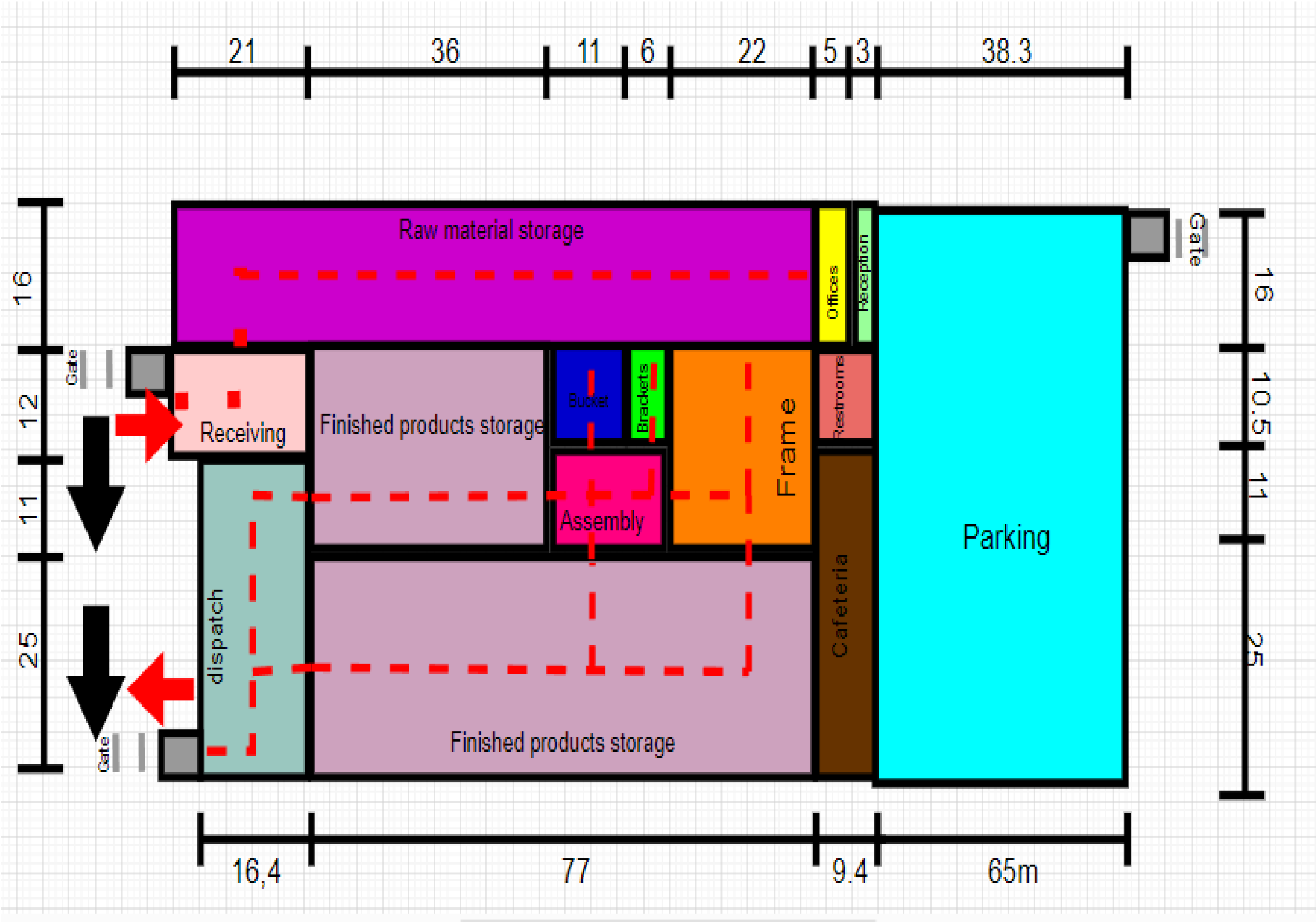

Figure 27:Alternative2

Alternative 3

In this alternative, there are 3 different gates, where 2 of them can serve as both entrances and exits for the staff and the receiving trucks, and the other gate for the dispatching trucks.

There are 2 different receiving areas as well as 2 different dispatching areas. The raw material storage is also split into two with the final product storage in the middle of them. This follows the operational process that can start at the two different storage areas. It will include the bucket and bracket departments at the one storage and the frame department at the other storage. The 2 separate parts will meet each other in the middle and go back again finally to the assemble department which is directly beside the final product storage area. This concludes to the operational process shaped into 2xU-shapes ending and beginning at approximately the same place. All the storage areas are above the manufacturing process with the dispatching and receiving areas and then the supporting facilities are below the manufacturing process with the parking that goes all around the supporting facilities and ends at the receiving areas.

The spitted storage areas, helps to use the space efficiently and minimises the material flow as the raw material will then be grouped at the different departments and can be transported more easily to the WIP stations. It ensures that the operational departments have efficient flow as the most important departments are grouped together and uses the space effectively as it is 2 u-shapes, so it minimises the distance travelled for the WIP materials as well as the material handling machines like the forklifts.

The 2 different receiving areas can help management to organise the receiving more efficiently as we receive a variety of different raw materials. They can split the specific types of receivals at the different receiving areas for the materials that would be more fitted for the specific storage area according to the type of operational department that is next to the storage. This improves control over the storage areas as well as the receiving process. It minimises traffic congestion as the trucks will receive the request beforehand what gate they should use.

The design supports the initial idea that all the operational processes end at the same point where the different parts that were made come together so that the final product can be assembled.

The separate dispatching areas can minimise traffic congestions but also split the orders so it can be handled more efficiently and quicker. A suggestion is that the one dispatch area is for the large orders and includes all of our own fleet at the one area and the other area only includes the other orders (smaller than 100) which also promotes the control over the system. In our design the 2 areas are the same sizes (300 m^2) but it can be determined how much of our daily/monthly orders are large vs. small and then adjustments can be made accordingly to the sizes of the areas.

The parking that goes around almost half of the facility with 2 different gates to decide on, it makes it more convenient for staff, visitors as well as receiving trucks as it is two different roads.

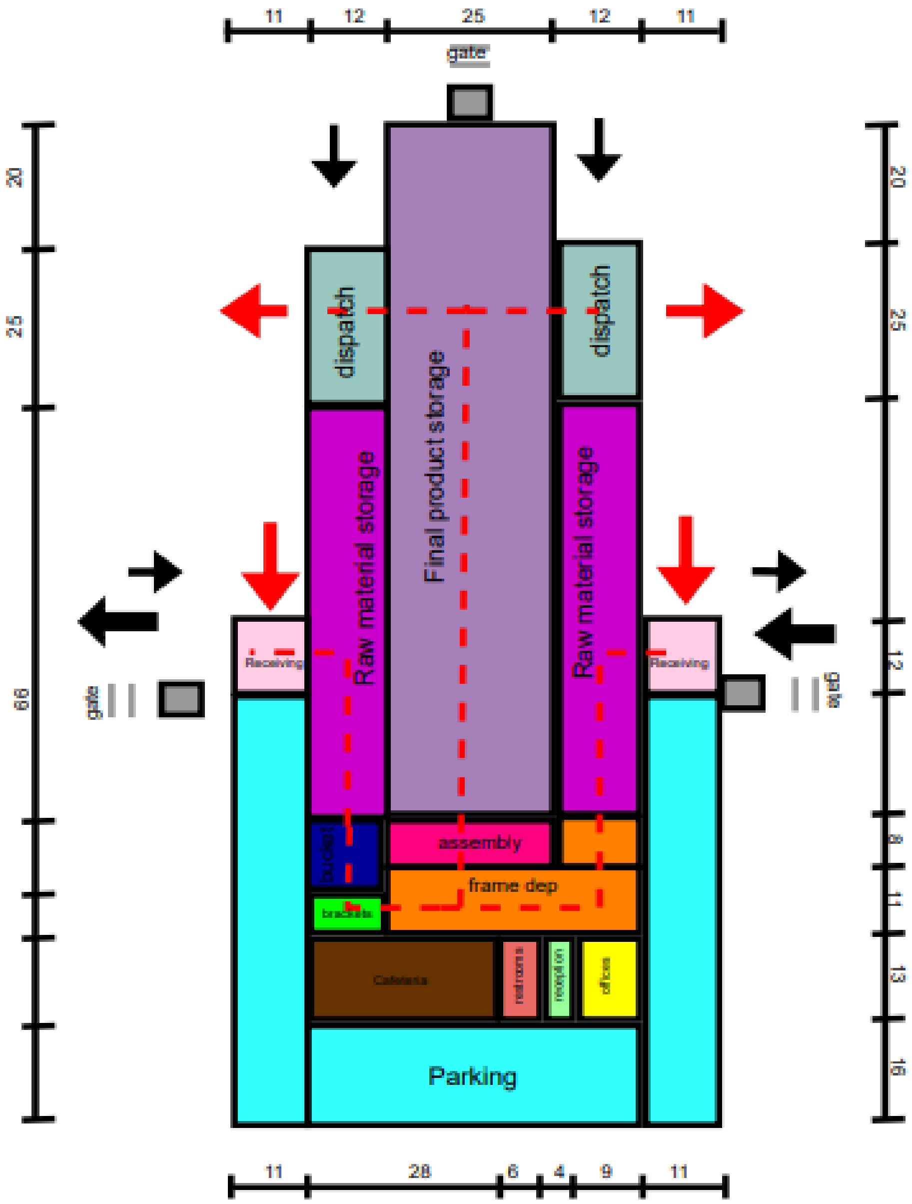

Figure 28:Alternative 3

8.Evaluation

We will be using a gap analysis to identify if each alternative design meets the requirements that is set for the new facility and choose the best design accordingly for the master facility design, by evaluating each design thoroughly.

Exceeds expectation	
Meets requirement	
Fail to meet requirement	

Table 24:GAP analysis

Criteria	Alternative			Comment
	1	2	3	
Promotes flow				Alternative 1's flow is not efficient as the reception blocks the flow between the raw material storage.
Space utilisation				Alternative 1 wastes a lot of space by not having a effective operational flow, and alternative 2 uses it the best as the storage areas are arranged logically to cooperate efficiently with the operational layout.
Minimal distance travelled between departments				In Alternative 1, the operational flow is almost around the whole plant, and some WIP materials also, must travel big distances which can lead to bottlenecks. The other alternatives operational materials travels min distance, but the cafeteria is far for the workers at the storage areas as well as the offices, would any of the staff want to view the storage areas.
Closeness of important departments				In Alternative 1, the bucket department is far away from the assembly department, which is important to be together. In both alternative 2 and 3 the reception is far from the dispatch and receiving areas.
Safety				Alternative 1 is not safe as the flow of some WIP materials are long distances which can cause possible safety risks like intersections with workers.
Quality control				Alternative 1's operational space makes it difficult to do quality control assurance.
Future expansion				All the alternatives provide space for potential 30 % expansion.
Promotes supply chain flow				Alternative 3 promotes it the best as it as 2 receiving and 2 dispatch areas.

Taking all of requirements into consideration it can be concluded that alternative 3 exceeded most of the requirements and is therefore the best design.

9.Final Design

The final design's total area spaces equate to 11 352 m^2. Note that this design took extra space for aisle space and forklift moving space. The design is done by a 2:1 ratio to show more detail, comparing to the alternative design layouts. Please refer to the separate pdf file for a higher quality version.

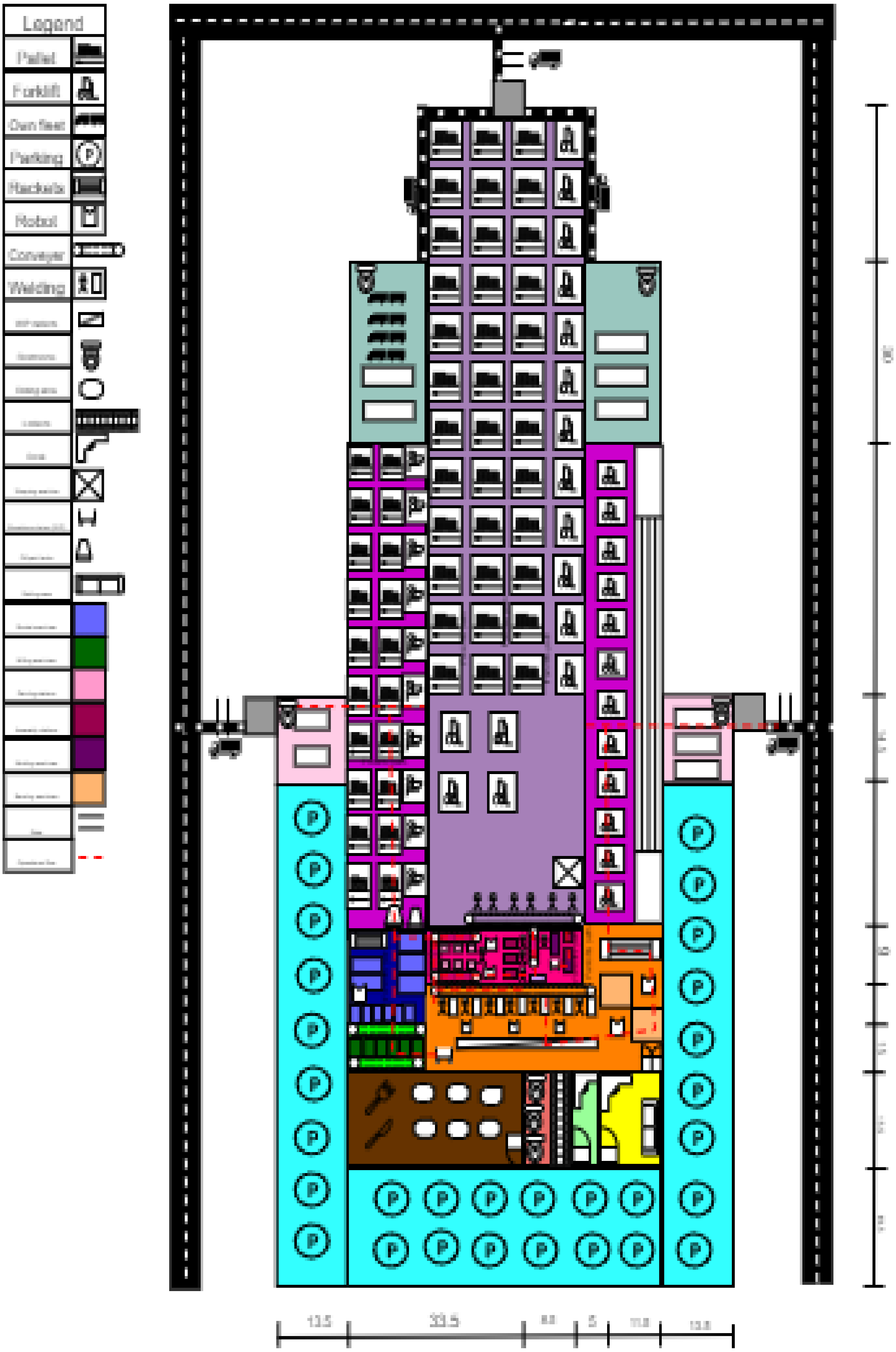

Figure 29:Master Facility Plan

10.Business case

10.1 Cost Estimates

10.1.1 Equipment costs

Table 25:Equipment costs

Equipment	Number of equipment	Cost of one	Total
Hydraulic press	11	R360 000	R3 960 000
tube bending machine	2	R105 000	R210 000
Milling machine	5	R700 000	R3 500 000
Welding machine	7	R5 000	R35 000
Injection molding machine	2	R1 700 000	R3 400 000
Painting guns	7	R850	R5 950
Conveyer	5	R300 000	R1 500 000
Forklifts	40	R1 200 000	R48 000 000
Plastic wrapper	1	R3 500 000	R3 500 000
Robot arms	6	R1 500 000	R9 000 000

10.1.2 Material Costs

Table 26:Material costs

Material	Number	Cost of one	Total
Steel sheets	20000	R130 000	R2 600 000 000
steel tubes	2680	R121 500	R325 620 000
wheels	20000	R450	R9 000 000
bolts, washers, nuts	20000	R3	R60 000
Oil	5000	R700 000	R3 500 000 000
Paint	5000	R33 000	R165 000 000
Plastic Pellets	300	R27 000	R27 000
Wrapping	74	R725	R53 650
Total			R6 599 760 650

10.1.3 Building Costs

To estimate the total cost for the facility we will assume a cost of R25 000/m^2 and R2000 per m^2 for parking. Therefore, our total will be:

$$Parking = 3411 * 2000 = R6822000$$

$$Departments = 7941 * 25000 = R198525000$$

10.2 Risks associated with implementation

When implementing a facility plan for a wheelbarrow manufacturing business, there are a few major concerns to consider. First, there is a danger that the project will go over budget. This might happen if the initial cost estimates are incorrect, the price of supplies rises unexpectedly, or there are delays in obtaining the essential licenses. Furthermore, any surprises during construction, such as identifying previously unknown concerns with the site, can result in additional expenditures. To avoid these financial problems, it's critical to plan of time, create a sound budget, and save some money just in case.

Another significant danger is that the construction process may impede the factory's normal operations. While production is in progress, any upgrades or expansions to the facility could disrupt everyday operations, slowing down output and perhaps raising safety issues. To reduce these delays, it is imperative that the building work be coordinated with the regular manufacturing schedule. Furthermore, inefficiencies may result from the new design's mismatch with the real requirements of the workforce and the production process, necessitating costly adjustments down the road. Workers and other stakeholders must be included in the planning phase to guarantee that the new facility satisfies their demands and promotes effective production to avoid this.